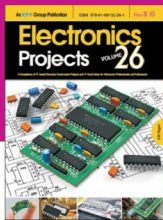




Electronics Projects Vol. 26 is the latest volume in the series, available in digital form.

It is a compilation of 21 construction projects and 71 circuit ideas published in Electronics For You magazine.



This collection of tested circuit ideas and construction projects in a handy volume would interest all classes of electronics enthusiasts—be they students, teachers, hobbyists or professionals!

Electronics Projects Vol. 26 is available on the following digital magazine stores:

- www.lulu.com
- www.magzter.com
- Createspace.com
- www.readwhere.com

Test and measurement for the IoT

In this era of the Internet of Things (IoT) and artificial intelligence, most applications are cloud based, open, modular and programmable. And the industry majors are working on the smart city project as part of Digital India initiative. Wireless equipment are now all-in-one solutions that use connectivity formats such as ZigBee, Bluetooth Low Energy, WiMAX and GPRS/EDGE/GSM.

The bandwidth of wireless test equipment has increased from 1Mbps to 10Gbps. The more the bandwidth, the fewer the chances of data loss and higher the chances of data retention. The functioning is in real time, so it is easy to capture the data and signals. Moreover, as the speed is high, continuous buffering is possible without loss. Results are displayed in PDF and JPEG formats.

Students working on smart-city projects need real-time data and extensive test and measurement techniques to get near-perfect results. Smartphone apps help them directly derive data from the test equipment and prepare reports in real time.

Analyser for optical fibre technology

Neeraj Dhawan, owner of Invas Technologies, shares that as the government is pushing the installation of optical fibre network across the country, research institutes are also training their students with test equipment for measuring the performance and capability of optical fibres.

Chromatic dispersion analyser is one such instrument that tests optical fibre channels in parallel. Institutes like IIT Kharagpur are completely focused on optical fibre test. Such institutes have tie-ups with the government to carry out fibre quality tests in their laboratory.

Chromatic dispersion analyser has been designed to test whether 100Gbps capacity can be carried in a single channel of optical fibre.

Portable spectrum analysers and logic analysers

Traditional spectrum analysers and logic analysers have also been replaced with portable equipment, which are easily accessible to researchers and students.

Dhanabal says that accessing FM radio waves, finding errors in them, changing the algorithms and coding was never better before portable equipment.

Comeback of analogue oscilloscopes

"Although educational institutes are moving towards smarter approach, conventional analogue oscilloscopes are not yet obsolete," says Manish Joshi, deputy CEO at Scientech Technology.

While a few manufacturers have ceased upgrading it, there are striking changes coming up, like TFT displays now showing voltage/division and time/division. Also, the working frequency of oscilloscopes has more than doubled the existing 20MHz.

Suman Murali Krishnan, a research associate from IIT Delhi, computer science engineering department, informs that they use a digital oscilloscope with voltage/timing analysis and USB interface with PC for data acquisition and analysis and for current measurement. IIT Delhi lab is an interdisciplinary lab with both electronic and mechanical equipment.

To sum up

In the future, experts expect to have a library for equipment as well. It would be similar to a book-library where the students will be able to borrow an equipment and return it after their work completes. It is an out-of-the-box concept. Also, students will have a forum to discuss these equipment. This smart approach will motivate students and researchers and help them develop skillset for the industry. **EFY**